

Field map for one patent record. See Sample Document #1 for a live JSON excerpt (~2-4 KB typical; claim\_1 may be longer).

Delivery unit

| Field          | Type   | Description                                                                                                 |
|----------------|--------|-------------------------------------------------------------------------------------------------------------|
| 1 order record | –      | One utility patent = one JSON object. Package tiers delivered as JSON array or JSONL (one object per line). |
| \$schema       | string | Fixed per delivery: "https://mirelia.site/llms.txt".                                                        |
| schema_version | string | Catalog label: "mirelia-patent-record-1.0" (same contract as \$schema).                                     |

Root – patent identity

| Field        | Type   | Description                                     |
|--------------|--------|-------------------------------------------------|
| patent_id    | string | Publication id, e.g. US12576658B2 (unique key). |
| appl_no      | string | Application number from USPTO XML.              |
| country_code | string | Issuing authority, e.g. "US".                   |
| title        | string | Invention title (English).                      |
| abstract     | string | Patent abstract text.                           |
| claim_1      | string | Full text of primary independent claim.         |

Provenance & legal metadata

| Field            | Type     | Description                                     |
|------------------|----------|-------------------------------------------------|
| source_bundle    | string   | Bulk source label, e.g. "USPTO".                |
| source_hash      | string   | SHA-256 of original USPTO XML for the record.   |
| onchain_evidence | string   | Optional blockchain tx id (proof of existence). |
| legal_status     | string   | Legal status when available.                    |
| family_id        | string   | Simple patent family id when available.         |
| filing_date      | date     | Application filing date (ISO).                  |
| publication_date | date     | Grant/publication date (ISO).                   |
| r2_url           | string   | CDN URL to this JSON artifact.                  |
| created_at       | datetime | Record creation timestamp (UTC).                |
| updated_at       | datetime | Last update timestamp (UTC).                    |

Parties & citation graph

| Field           | Type     | Description                          |
|-----------------|----------|--------------------------------------|
| assignee        | string[] | Corporate assignees.                 |
| inventor        | string[] | Inventor names.                      |
| priority_claims | string[] | Priority claim references.           |
| cited_patents   | string[] | Examiner/applicant patent citations. |
| npl_citations   | string[] | Non-patent literature citations.     |

Classification (CPC / IPC)

| Field          | Type     | Description                                        |
|----------------|----------|----------------------------------------------------|
| primary_cpc    | string   | Primary Cooperative Patent Classification section. |
| secondary_cpcs | string[] | Additional CPC sections.                           |
| ipc_classes    | string[] | IPC main classifications from XML.                 |

AI-enriched business & technology fields

| Field           | Type     | Description                                                             |
|-----------------|----------|-------------------------------------------------------------------------|
| importance_p    | int      | Proprietary score 1-100 (innovation, barrier, market impact).           |
| biz_value_prop  | string   | Problem, solution, and business advantage (RAG-friendly, no metaphors). |
| biz_target_ind  | string[] | Target industries for licensing/product fit.                            |
| attr_tech_stack | string[] | Core technology tags.                                                   |

attr\_performance[] – quantified specs

| Field          | Type          | Description                                            |
|----------------|---------------|--------------------------------------------------------|
| metric         | string        | Named performance/spec metric from claims or abstract. |
| value_original | number string | Value as disclosed in the patent.                      |
| unit_original  | string        | Unit as disclosed, e.g. "psi", "time".                 |
| value_defacto  | number null   | SI-normalized value when computed in pipeline.         |
| unit_defacto   | string null   | SI unit after normalization.                           |

Commerce & integrity

| Field        | Type     | Description                                                  |
|--------------|----------|--------------------------------------------------------------|
| package_tags | string[] | Catalog package tags (CPC bundles).                          |
| sales_count  | int      | Purchase count for this record.                              |
| _metadata    | object   | source_bundle, processed_at, source_hash (processing audit). |
| _signature   | string   | HMAC-SHA256 over canonical JSON (tamper detection).          |

Packages group 10-1,000 patents by CPC category via package\_tags. AI fields are inferred from title, abstract, and claim\_1. MCP discovery: list\_available\_categories → get\_package\_sample → verify\_crypto\_payment\_and\_deliver.